

MTH 310/520 Graph Theory: Quiz 1

Solution

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Total: 10 points

Given a graph G , an *ear decomposition* of a graph G is a collection of paths $P_1, \dots, P_m \subseteq G$ and a cycle $P_0 \subseteq G$, such that $\cup_{i=0}^m P_i = G$, and for any $1 \leq i \leq m$, the end points of P_i lie in $V(\cup_{j=0}^{i-1} P_j)$ and no interior point of P_i lies in $V(\cup_{j=0}^{i-1} P_j)$. The cycle and paths $P_0, \dots, P_m$ are referred to as *ears*. In this quiz, we shall prove the following result that $\kappa(G) \geq 2 \iff G$ admits an ear decomposition.

1) In this question, we show that $\kappa(G) \geq 2 \implies G$ admits an ear decomposition.

a) Show that $\kappa(G) \geq 2$ implies G has a cycle.

b) Define P_0 to be a cycle obtained in Q1.a. Now assume that you have already constructed the ears $P_0, P_1, \dots, P_i$. Define $A := V(\cup_{j=0}^i P_j)$, and $B := V(G) \setminus A$. If $A = V(G)$ does $P_0, \dots, P_i$ necessarily constitute an ear decomposition? If not, show that one can add more ears to complete the ear decomposition.

c) Now, suppose $A \subsetneq V(G)$. Show that you can always add one more ear P_{i+1} .

[Hint: Observe that $B \neq \emptyset$. Can you suitably use Menger's theorem between A and some component of $G[B]$ to construct P_{i+1} ?]

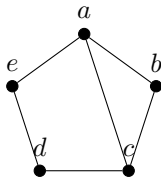
[1+2+4=7 points]

Solution:

a) From the inequality $\delta(G) \geq \kappa(G)$, we get $\delta(G) \geq 2$. Hence G has a cycle.

b) No, since all edges might not be covered by the decomposition. For the given C_5 with a chord, $P_0 = C_5$ satisfies the condition in b) isn't the full decomposition of the graph.

For $e \notin E(\cup_{j=0}^i P_j)$, $P_e = e$ is an ear since its endpoints lie in A and its interior points don't lie in A , add P_e to the decomposition, to complete the ear decomposition.



- c) Let $v \in B$, look at the v -A fan in the original graph, Since $\kappa(G) \geq 2$, there exist a v -A fan of size 2. (Fan version of Menger's Theorem). Let the path and their respective endpoints be $P_1, P_2 | v_1, v_2$ then by joining the two paths like $v_1 - v - v_2$ we get an ear as its a path (since P_1 and P_2 are internally disjoint), the endpoints lie in A , and no interior point lies in A (since its a v -A fan).

Hence we can always add 1 more ear if $B \neq \emptyset$.

- 2) In this question, we show the reverse direction, i.e., if G admits an ear decomposition then $\kappa(G) \geq 2$.

- a) Let $P_0, \dots, P_m$ be an ear decomposition for G . Then for any $0 \leq i \leq m$, show that $G[V(\cup_{j=0}^i P_j)]$ is connected.
- b) For any $v \in V(G)$, define $\text{ear}(v)$ to be the least index i such that $v \in V(P_i)$. Show using Q2.a) that vertices in $V(\cup_{i=0}^{\text{ear}(v)} P_i) \setminus \{v\}$ lie in the same component of $G \setminus \{v\}$.
- c) Next, show that for any $k > \text{ear}(v)$ vertices in $V(\cup_{i=0}^k P_i) \setminus \{v\}$ lie in the same component of $G \setminus \{v\}$ provided vertices in $V(\cup_{i=0}^{k-1} P_i) \setminus \{v\}$ lie in the same component of $G \setminus \{v\}$.
- d) Argue using the above that $\kappa(G) \geq 2$.

[1+1+0.5+0.5=3 points]

Solution:

- a) We will do induction on i

Base case: P_0 is a cycle and hence its connected by definition

Inductive hypothesis: Let $G[V(\cup_{j=0}^i P_j)]$ be connected for all $i \leq k$

Inductive step: For $i = k + 1$,

Case 1: $v, u \in V(\cup_{j=0}^{i-1} P_j)$ are connected via the inductive hypothesis

Case 2: $u, v \in P_i$, are connected since they lie on a path.

Case 3: Let v_e be the endpoint of the ear P_i then $u \in P_i$ and $v \in V(\cup_{j=0}^{i-1} P_j)$ are connected as there is a path from $v - v_e$ and $v_e - u$. Joining them gives us walk which then gives us a $u - v$ path.

Hence $G[V(\cup_{j=0}^i P_j)]$ is connected

- b) Let $v \in V(G)$ and let $i = \text{ear}(v)$. Then $v \in V(P_i)$

By Q2.a), $G[V(\cup_{j=0}^i P_j)]$ is connected. We show that removing v does not disconnect this subgraph. Again by Q2.a) $V(\cup_{j=0}^{i-1} P_j)$ is connected. Since the endpoints of P_i lie in $V(\cup_{j=0}^{i-1} P_j)$, every vertex in $P_i \setminus \{v\}$ has a path to $V(\cup_{j=0}^{i-1} P_j)$ avoiding v and following from this any $a, b \in P_i \setminus \{v\}$ are also connected (by the concatenated path)

Hence they lie in the same component of $G \setminus \{v\}$.

- c) Let $S_j = V(\cup_{i=0}^j P_i) \setminus \{v\}$. Assume S_{k-1} lies in a single component of $G \setminus \{v\}$. Since $k > \text{ear}(v)$, v is not an internal vertex of P_k . At least one endpoint of P_k , say u , must lie in S_{k-1} (v can be one of the endpoints). Every vertex in $P_k \setminus \{v\}$ is connected to u , and hence S_{k-1} along P_k without passing through v . Thus, $S_k = S_{k-1} \cup (P_k \setminus \{v\})$ also lies entirely in the same component of $G \setminus \{v\}$.
- d) For any $v \in V(G)$, Q2.b and repeated application of Q2.c (for $k = \text{ear}(v) + 1 \dots m$) imply that $V(\cup_{i=0}^m P_i) \setminus \{v\} = V(G) \setminus \{v\}$ is connected. Since removing any arbitrary vertex v leaves G connected, G has no cut-vertex. Therefore, $\kappa(G) \geq 2$.